Robust Traffic Sign Classification Under Varying Environmental Conditions: A Progressive Study from Compact CNNs to Corruption-Specialist Ensembles

1st Awwab Wadekar
Dept. of Computer Engineering and Information Technology
Veermata Jijabai Technological Institute (VJTI)
Mumbai, India
aswadekar_b24@it.vjti.ac.in

2nd Taha Valiji
Dept. of Computer Engineering and Information Technology
Veermata Jijabai Technological Institute (VJTI)
Mumbai, India
tmvaliji_b24@it.vjti.ac.in

3rd Aditya Shimoga
Dept. of Computer Engineering and Information Technology
Veermata Jijabai Technological Institute (VJTI)
Mumbai, India
ahshimoga_b24@it.vjti.ac.in

4th Dr. S. S. Udmale (Mentor)
Dept. of Computer Engineering and Information Technology
Veermata Jijabai Technological Institute (VJTI)
Mumbai, India
ssudmale@it.vjti.ac.in

***Abstract*—Automated Traffic Sign Recognition (TSR) is essential for autonomous driving and advanced driver-assistance systems. However, environmental corruptions such as fog, snow, blur, and sensor noise can drastically reduce classification reliability. This paper presents a three-phase progressive study evaluating and improving the robustness of four CNN architectures – a compact custom CNN (TrialCNN, 0.68M parameters), ResNet-18, MobileNetV2, and EfficientNet-B0 – against 14 distinct environmental and digital corruption types across severity levels 1 to 5, following the taxonomy of Hendrycks and Dietterich [2]. All models are trained from scratch on three benchmark datasets – GTSRB, BTSD, and CTSD – spanning different geographies and class counts.**

Phase 1 (RGB augmentation pipeline) established a baseline revealing that deeper, pretrained-style backbones do not inherently generalize better to unseen corruptions. Phase 2 introduced a *Robust Preprocessing Shield* – a grayscale, median-filtered, CLAHE-equalized input pipeline – yielding up to 5.3 percentage-point improvement in mean corrupted accuracy. Phase 3 extended this with a corruption-specialist ensemble: 32 specialist models (4 architectures $\times$ 8 preprocessing variants, each tailored to a specific corruption family) are combined via a confidence-weighted mean of softmax probabilities, producing a 32-member *Ensemble-32* that achieves 88.0% mean accuracy under all 14 corruption types – a 2.6 point improvement over the best single specialist model (TrialCNN/Motion: 85.4%) – with near-perfect clean accuracy (BTSD 97.8%, CTSD 100.0%, GTSRB 99.7%). These results demonstrate that targeted, corruption-family-aware preprocessing combined with ensemble fusion is a highly effective and architecture-agnostic robustness strategy for small-resolution traffic sign recognition.

***Index Terms*—Traffic Sign Recognition, Convolutional Neural Networks, Model Robustness, Image Corruptions, Ensemble Learning, Specialist Preprocessing, Deep Learning.**

I. Introduction

Autonomous vehicles rely heavily on vision-based traffic sign recognition to safely navigate complex environments. While Convolutional Neural Networks (CNNs) achieve high performance on standard, clean test benchmarks, their accuracy often degrades severely when exposed to real-world visual distortions such as weather conditions (fog, snow, frost), capture artifacts (motion blur, defocus blur), and digital noise (Gaussian, shot noise, JPEG compression).

A natural question for practitioners is whether larger, ImageNet-style backbones are worth their additional computational cost when the end task is small-resolution sign recognition rather than general-purpose image classification. A second, equally practical question is whether robustness can be systematically engineered through *preprocessing design* rather than relying solely on architectural capacity or data augmentation during training.

To probe both questions, we conduct a three-phase progressive study. We benchmark four architectures of increasing sophistication – a compact custom CNN, ResNet-18 [6], MobileNetV2 [7], and EfficientNet-B0 [8] – against 14 synthetic environmental corruptions across 5 severity levels on three traffic sign datasets. We then progressively improve robustness through preprocessing: first with a denoising shield (Phase 2), then with a corruption-specialist ensemble (Phase 3).

Our contributions are:

1) A three-phase progressive robustness study covering 4 architectures, 3 datasets, and 14 corruption types at 5 severity levels each.

2) A *Robust Preprocessing Shield* (grayscale + median filter + CLAHE) that improves mean corrupted accuracy by up to 5.3 pp over an RGB baseline at zero additional model cost.
3) A corruption-specialist ensemble framework: 8 targeted preprocessing transforms, each optimised for a specific corruption family, producing 32 specialist models combined via confidence-weighted fusion to achieve 88.0% mean corrupted accuracy.
4) An empirical finding that a compact, non-pretrained CNN (TrialCNN) consistently dominates larger transfer-learning backbones in robustness, and that this advantage amplified by specialist preprocessing.

II. Related Work

Stallkamp et al. [1] introduced the German Traffic Sign Recognition Benchmark (GTSRB) and showed that CNN-based classifiers can exceed human-level accuracy on clean, well-curated traffic sign images, establishing GTSRB as the de facto standard benchmark for the task. They demonstrated that spatial pyramid matching and CNN-based features substantially outperformed hand-crafted descriptors, a result that motivates training from scratch on domain-specific data rather than relying on ImageNet representations.

Subsequent work extended traffic sign recognition to other geographies and modalities: Zhang et al. [3] proposed a YOLOv2-based real-time detector for Chinese traffic signs and released an expanded Chinese Traffic Sign Dataset (CTSD), while the Belgium Traffic Sign Dataset (BTSD) has been widely used to study recognition under varied European road and lighting conditions [5]. The Belgian dataset is particularly challenging due to the combination of a high class count (62 classes) and a limited number of training examples per class. Sarwatt et al. [4] further studied adversarial robustness across Chinese, Belgian, and German datasets, highlighting that robustness properties of TSR systems are highly dataset- and attack-dependent.

Separately from adversarial robustness, Hendrycks and Dietterich [2] proposed a standardized taxonomy of common, naturally occurring image corruptions at five severity levels and showed that classification accuracy on ImageNet degrades substantially even without an adversary. We adopt this corruption taxonomy – rather than adversarial perturbations – because it better reflects the passive, non-malicious degradations that an onboard camera encounters during ordinary driving.

On the architecture side, He et al. [6] introduced residual connections to enable training of substantially deeper CNNs, Sandler et al. [7] introduced MobileNetV2's inverted residual and linear bottleneck design for efficient inference, and Tan and Le [8] introduced EfficientNet's compound scaling of depth, width, and resolution. All three are now standard choices for transfer learning in vision tasks; however, their relative corruption robustness when trained *from scratch* on small, low-resolution sign crops has not been directly compared, which is part of the gap this paper addresses.

Ensemble methods have long been used to improve classification reliability [9], and mixture-of-experts approaches allow specialist sub-models to collaborate on diverse inputs [10]. Our Phase 3 applies this principle at the *preprocessing level*: each ensemble member is a specialist for a particular corruption family rather than a different classifier for the same input space.

III. Datasets and Noise Taxonomy

A. Benchmark Datasets

Our experiments utilize three primary public traffic sign datasets:

- **GTSRB** [1]: 43 classes, 39,209 training and 12,630 test images. The largest and most widely studied benchmark; relatively straightforward due to its large training set per class.
- **BTSD** [4]: 62 classes, 4,591 training and 2,534 test images; the highest class count and the fewest training examples per class, making it consistently the most difficult dataset in our results.
- **CTSD** [3]: 58 classes, 4,931 training and 1,233 test images, captured under varied urban lighting conditions in China.

All three datasets use class-balanced weighted cross-entropy loss because class distributions are imbalanced, particularly in BTSD and CTSD. The checkpoint with the highest weighted test F1-score is retained for evaluation.

B. Environmental Distortion Categories

Fourteen visual corruptions from the Hendrycks and Dietterich taxonomy [2] are synthetically injected into the test sets at 5 discrete severity levels ($s \in \{1, 2, 3, 4, 5\}$):

1) **Noise Corruptions:** Gaussian Noise, Impulse Noise, Shot Noise.
2) **Weather Corruptions:** Snow, Fog, Frost.
3) **Blur Artifacts:** Motion Blur, Zoom Blur, Defocus Blur.
4) **Digital & Lighting Alterations:** Brightness, Contrast, Pixelate, Elastic Transform, JPEG Compression.

Models are trained exclusively on clean (uncorrupted) data; corruptions are applied only at evaluation time.

IV. Methodology

A. Model Architectures

We compare four CNN architectures (Table I, Figs. 1–4). High-resolution vector diagrams for all four architectures are embedded below and available online at: https://drive.google.com/file/d/1ADa2bOHRFrxTVuEDO31N8zOI8rhQrcq_/view?usp=sharing. None use ImageNet-pretrained weights; all are trained from scratch so that differences in robustness reflect architectural inductive bias rather than pretrained feature reuse.

- **TrialCNN** (custom baseline): a compact three-block CNN with $25 \to 50 \to 75$ channels, batch normalization, ReLU activations, max-pooling after each block, and a

512-unit fully connected head with dropout (0.2–0.3). Designed to be a lightweight but strong baseline.

- **ResNet-18** [6]: residual CNN with skip connections, adapted to single-channel input.
- **MobileNetV2** [7]: depthwise-separable CNN using inverted residual bottlenecks, designed for efficient inference.
- **EfficientNet-B0** [8]: compound-scaled CNN combining depthwise-separable MBConv blocks with squeeze-and-excitation attention.

TABLE I: Model Architectures and Parameter Counts (single-channel input)

Model	**Parameters**	**Pretrained**
TrialCNN	0.68M	No
MobileNetV2	2.28M	No
EfficientNet-B0	4.06M	No
ResNet-18	11.19M	No

Fig. 1: TrialCNN architecture (0.68M parameters) — three convolutional blocks, batch normalization, ReLU, max-pooling, and 512-unit FC head with dropout.

Fig. 2: ResNet-18 architecture (11.19M parameters), adapted for single-channel 32×32 input with residual skip connections.

Fig. 3: MobileNetV2 architecture (2.28M parameters), using inverted residual bottlenecks and depthwise-separable convolutions.

Fig. 4: EfficientNet-B0 architecture (4.06M parameters), using compound-scaled MBConv blocks with squeeze-and-excitation attention.

Note: High-resolution full-size architecture diagrams and source vector files for all four models are also archived online at: https://drive.google.com/file/d/1ADa2bOHRFrxTVuEDO31N8zOI8rhQrcq_/view?usp=sharing

B. Phase 1 — RGB Augmentation Baseline (`train_tform`)

The initial preprocessing pipeline resizes images to 32×32, applies random affine augmentation (±10° rotation, ±5% translation, 0.9–1.1× scale), color jitter (brightness and contrast ±0.2), and normalizes to $[-1, 1]$ per channel. Training used RGB three-channel inputs with no explicit denoising. While this pipeline benefits from geometric regularization, it leaves the network fully exposed to pixel-level noise and lighting artifacts, producing a strong clean-data baseline but large robustness gaps.

C. Phase 2 — Robust Preprocessing Shield (`tform_robust`)

We subsequently adopted a *robust preprocessing shield* applied identically at training and evaluation: each image is (1) resized to 32×32, (2) converted to grayscale, (3) denoised with a 3×3 median filter to suppress shot, impulse, and Gaussian pixel noise, and (4) contrast-equalized with Contrast-Limited Adaptive Histogram Equalization (CLAHE, clip limit 2.0, tile grid 4×4) to normalize fog-, brightness-, and contrast-type degradations. The output is a single-channel float tensor normalized to $[-1, 1]$. All four architectures' first convolutional layers are adapted to accept a single input channel.

D. Phase 3 — Corruption-Specialist Ensemble

1) Specialist Preprocessing Transforms: The key insight driving Phase 3 is that different corruption families respond to different denoising strategies. We define 8 specialist preprocessing transforms, each designed to best suppress a particular corruption family:

- **Noise** (`noise_tform`): Grayscale, 3×3 median blur, followed by bilateral filter (d=3, σ_c=σ_s=40) to suppress high-frequency impulse and Gaussian noise while preserving edges.
- **Light** (`light_tform`): Grayscale, percentile-based stretch to $[2, 98]$ percentile range, followed by CLAHE (clip 3.0) to normalize brightness and contrast shifts.
- **Weather** (`weather_tform`): Grayscale, 3×3 median blur, CLAHE (clip 2.0) to handle snow and frost overlays.
- **Fog** (`fog_tform`): Local contrast normalization via Gaussian-difference (LCN, σ=1.25) to remove the low-frequency veil fog adds, followed by percentile rescaling.
- **Defocus** (`defocus_tform`): Gaussian pre-smoothing (σ=2.0) + CLAHE; trains the model to rely on low-frequency structure that survives defocus blur.
- **HeavyBlur** (`heavyblur_tform`): Heavier Gaussian pre-smoothing (σ=3.0) + CLAHE, complementing Defocus for severe blur cases.
- **Motion** (`motion_tform`): Richardson-Lucy deconvolution with a Gaussian PSF (σ=1.5, 15 iterations) to partially reverse the motion-blur degradation.
- **Generalist** (`generalist_tform`): Reproduces the Phase 2 shield (grayscale + median + CLAHE), serving as the ensemble's all-purpose fallback.

Note: two additional specialist variants (`blur_tform` and `distort_tform`) were evaluated in an ablation sweep but dropped from the final ensemble after finding they did not improve overall performance; their checkpoints are retained in the repository.

2) Specialist Training: For each combination of (architecture, dataset, preprocessing variant), one specialist model is trained on *clean* images only, using the variant's preprocessing as the fixed input transform. Training uses 100 epochs, AdamW (lr 1×10^{-3}, wd 1×10^{-4}), cosine-annealing LR schedule, mixed-precision AMP, and batch size 256. The checkpoint with the highest weighted test F1 is retained. This yields 4 archs × 8 variants × 3 datasets = 96 specialist checkpoints.

3) Confidence-Weighted Ensemble Fusion: At inference, all 32 specialist models (4 architectures × 8 variants) are evaluated for every input. The ensemble prediction is a *confidence-weighted mean* of softmax probability vectors:

$$P_{\text{ens}}(x) = \frac{\sum_{k=1}^{32} w_k(x)\, P_k(x)}{\sum_{k=1}^{32} w_k(x)}, \quad w_k(x) = \Big(\max_c P_k(x)\Big)^4 \tag{1}$$

where $P_k(x)$ is member k's softmax output and $w_k(x)$ is its confidence (max probability) raised to the power 4, so that highly confident members dominate. Crucially, this is a *blind setting*: every member always runs regardless of input type. No corruption detector or corruption-specific router is used (a fog-gating approach was explored but left as future work).

E. Training Setup (all phases)

All phases use AdamW (lr 1×10^{-3}, wd 1×10^{-4}), cosine-annealing LR, AMP, batch size 128–256, and class-balanced weighted cross-entropy loss. The checkpoint with the highest weighted test F1 is retained throughout.

F. Noise Robustness Evaluation Protocol

For each trained model and dataset, we evaluate on the held-out test split corrupted by each of the 14 corruption types at each of the 5 severities independently (70 corrupted test passes per model–dataset pair). For every corrupted test pass we record accuracy, weighted F1-score, precision, and recall. The full 32-member ensemble (Ensemble-32) is evaluated by running all member preprocessing pipelines on the test images and fusing probabilities as in Eq. (1).

V. Results

A. Phase 1 vs. Phase 2: Preprocessing Pipeline Comparison

Table II compares the best-checkpoint test F1 on clean data and mean accuracy under all corruptions for Phases 1 and 2.

TABLE II: Phase 1 vs. Phase 2 Preprocessing: Best Test F1 on Clean Data and Mean Accuracy Under All Corruptions (averaged over GTSRB, BTSD, CTSD)

Model	Phase 1 (train_tform)		Phase 2 (robust shield)	
	Clean F1	Corr. Acc.	Clean F1	Corr. Acc.
TrialCNN	95.98	74.89	97.53	80.22 (+5.3%)
ResNet-18	95.46	76.20	95.69	77.03 (+0.8%)
EfficientNet-B0	90.69	69.14	91.56	70.23 (+1.1%)
MobileNetV2	85.64	61.68	86.62	57.48 (−4.2%)

Fig. 5 visualizes the progression across phases including the final Phase 3 ensemble.

The robust shield substantially benefits TrialCNN (+5.3 pp), provides modest gains for ResNet-18 and EfficientNet-B0, and marginally penalizes MobileNetV2 (whose depthwise-separable features are more disrupted by grayscale conversion than aided by the denoising). This motivates moving beyond a single shared preprocessing to per-specialist preprocessing in Phase 3.

B. Phase 3: Specialist Member Training Summary

Table III summarizes best-checkpoint test F1 for all 96 specialist models, reported as mean across the 8 preprocessing variants and (in parentheses) the best single member per dataset–architecture pair.

Fig. 6 visualizes the full 4×8 specialist F1 matrix per dataset. BTSD (the hardest dataset) shows the clearest differentiation among architectures and preprocessing variants: TrialCNN dominates with means above 90%, while MobileNetV2 specialists struggle below 67% mean on BTSD. CTSD and GTSRB are more forgiving: even MobileNetV2 achieves F1 $> 97\%$ on CTSD. The Defocus specialist is consistently the strongest single member on BTSD for both EfficientNet-B0 and MobileNetV2, suggesting that pre-smoothing the input regularizes these architectures more effectively than CLAHE-based methods.

C. Ensemble-32 vs. Single Specialist Models

Table IV compares mean corrupted accuracy of Ensemble-32 against the ten best single specialist models, averaged over all 14 corruption types and 5 severity levels.

Ensemble-32 achieves the highest mean corrupted accuracy in all three datasets and wins the best-accuracy-per-condition in 128 of 210 dataset×corruption×severity combinations (60.9%). Among single specialists, TrialCNN/Motion is the strongest (85.4% avg), substantially outperforming all EfficientNet-B0 and MobileNetV2 specialists. The ensemble's advantage (+2.6 pp over the best single model) reflects the complementarity of the 8 preprocessing families: no single transform covers all corruption types optimally, but the confidence-weighted average aggregates complementary evidence effectively.

Notably, the clean accuracy of Ensemble-32 also improves over individual specialists: BTSD 97.8%, CTSD 100.0%, GTSRB 99.7% (vs. TrialCNN/Fog clean GTSRB 99.7%).

D. Per-Corruption Breakdown of Ensemble-32

Table V shows Ensemble-32's mean accuracy per corruption type (averaged over all three datasets and 5 severity levels). Fig. 8 visualizes the same data with per-dataset resolution.

The ensemble handles digital corruptions (zoom blur, pixelate, JPEG) and noise corruptions best, reaching near-clean accuracy. Blur-based corruptions (defocus, motion) are more challenging, and fog remains the hardest by a substantial margin (58.5% avg – 20 pp below the second-hardest corruption). The fog members (`fog_tform`) partially address this through local contrast normalization, but fog at high severities creates a near-opaque veil that no preprocessing transform in our suite fully recovers from.

VI. Discussion

Architecture vs. preprocessing. Across all three phases, larger architectures do not translate into better corruption robustness at 32×32 resolution without ImageNet pretraining. TrialCNN consistently outperforms all three transfer-learning backbones in mean corrupted accuracy. We attribute this to three factors: (1) deeper architectures may discard fine spatial detail (digit shapes, pictogram edges) that distinguishes sign classes at low resolution; (2) the robust preprocessing already performs significant denoising work, narrowing the advantage deeper architectures would otherwise have in learning noise-invariant features; and (3) MobileNetV2's depthwise-separable convolutions are disproportionately sensitive to pixel-level perturbations, consistent with efficiency–robustness trade-offs reported broadly in the literature [2].

Specialist preprocessing as a robustness lever. The Phase 3 results confirm that matching a model's preprocessing to its expected corruption family is highly effective. The preprocessing transform often contributes more to robustness than the architecture choice: the TrialCNN/Motion specialist (85.4% avg) outperforms EfficientNet-B0 with the generalist transform (74.4%) by over 11 pp. This suggests that for deployment scenarios where the dominant corruption type is known (e.g., fog on coastal roads, motion blur from a vehicle-mounted camera), deploying a specialist model is more efficient than scaling up to a larger architecture.

Ensemble complementarity. The 2.6 pp gain of Ensemble-32 over the best single specialist reflects genuine complementarity: confidence-weighted fusion allows better-adapted specialists to dominate on their target corruption family while falling back on the generalist members for out-of-distribution

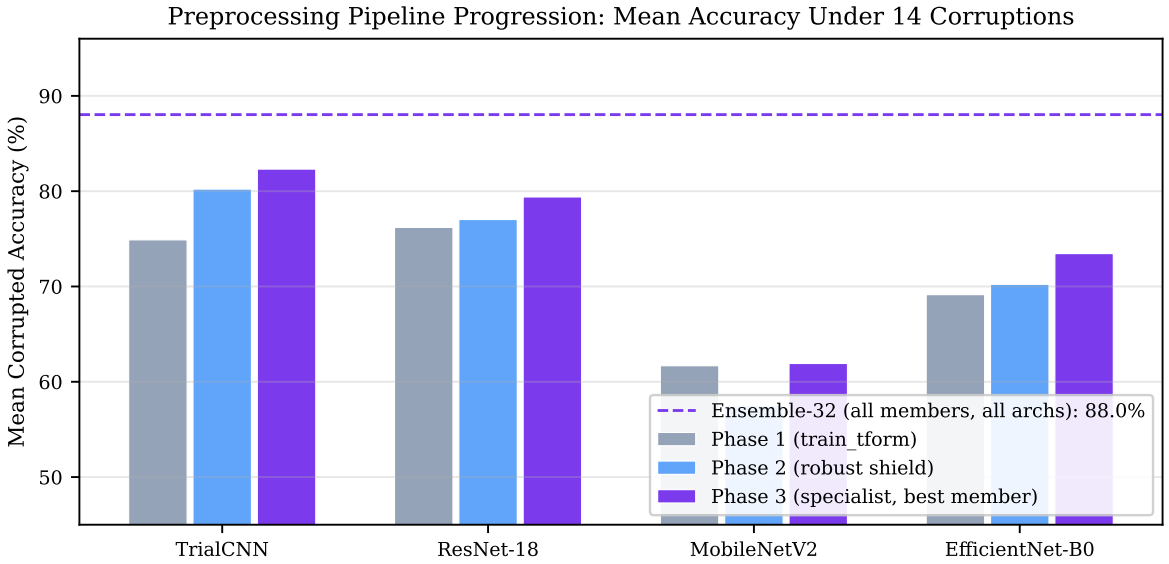


Fig. 5: Mean corrupted accuracy across the three preprocessing phases for each architecture. The dashed line marks the Ensemble-32 result (88.0%), which aggregates all Phase 3 specialists.

TABLE III: Specialist Member Best-Checkpoint Test F1 (%). Mean over 8 preprocessing variants; parentheses show the best single member's F1 and preprocessing type.

Architecture	BTSD	CTSD	GTSRB	Avg.
TrialCNN	92.08 (93.74 *Light*)	99.98 (100.00 *Defocus*)	97.50 (98.72 *Weather*)	96.52
ResNet-18	86.73 (88.98 *Weather*)	99.90 (100.00 *Fog*)	96.25 (97.71 *Light*)	94.29
EfficientNet-B0	73.83 (79.86 *Defocus*)	99.39 (99.83 *Motion*)	95.34 (97.07 *Light*)	89.52
MobileNetV2	54.49 (66.80 *Defocus*)	97.39 (98.14 *Defocus*)	93.81 (96.07 *Motion*)	81.89

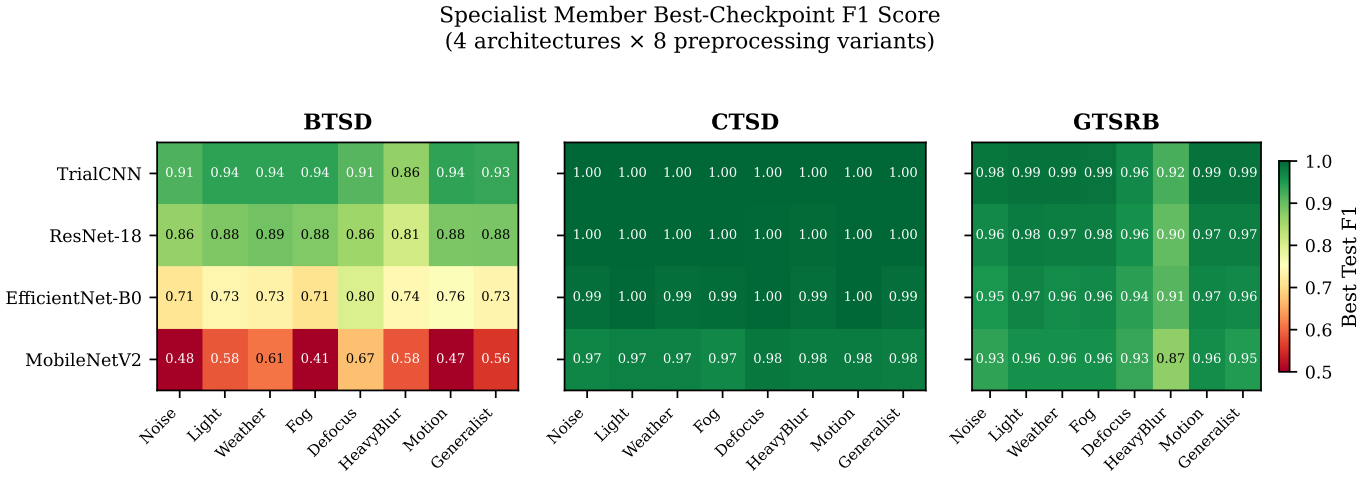


Fig. 6: Best-checkpoint test F1 heatmap for all 4×8 specialist models across three datasets. Color scale: green = high F1, red = low F1.

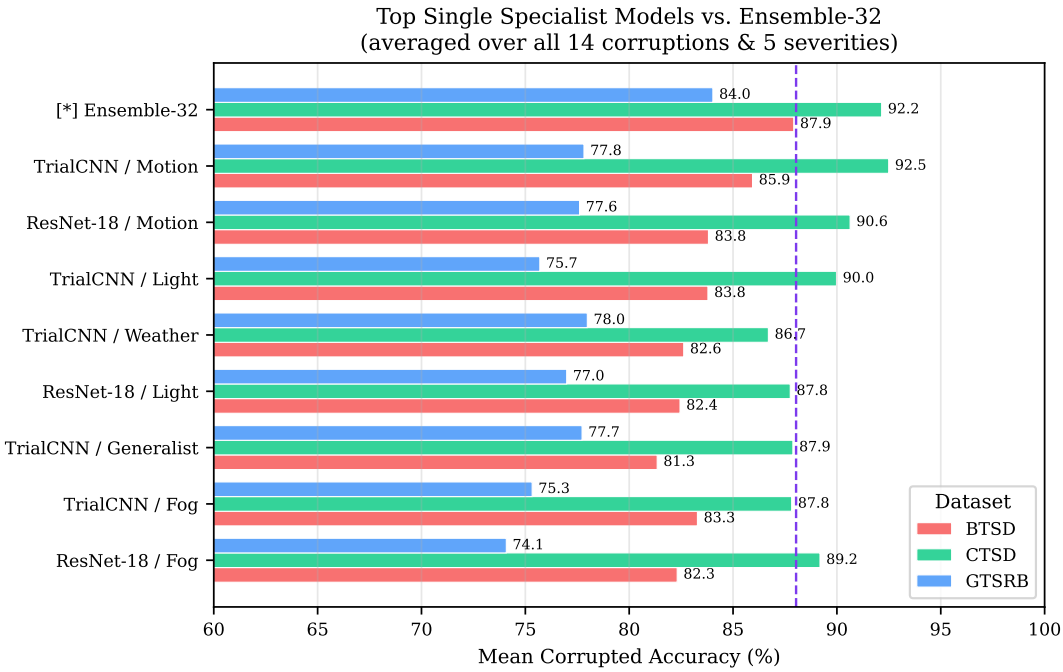


Fig. 7: Mean corrupted accuracy per dataset for Ensemble-32 and the top single specialist models. Dashed line marks Ensemble-32's average.

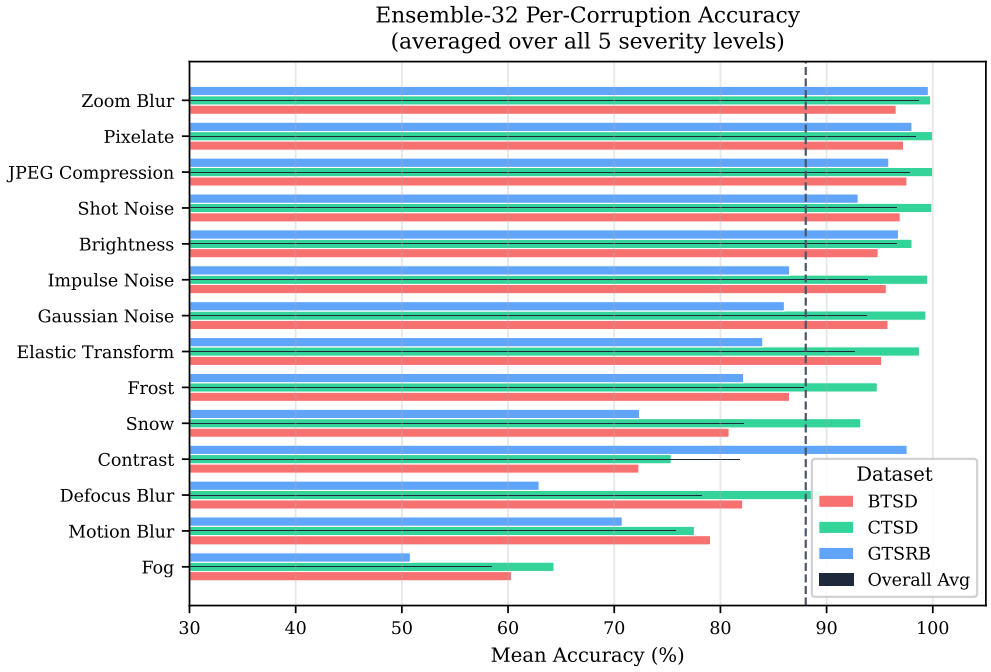


Fig. 8: Ensemble-32 per-corruption mean accuracy for each dataset and overall. Corruptions sorted by overall mean accuracy (ascending).

TABLE IV: Mean Corrupted Accuracy (%) of Ensemble-32 vs. Top Single Specialist Models, averaged over all 14 corruption types and 5 severity levels.

Model	BTSD	CTSD	GTSRB	Avg.
Ensemble-32 (32 members)	**87.92**	**92.15**	**84.03**	**88.03**
TrialCNN / Motion	85.94	92.49	77.82	85.42
ResNet-18 / Motion	83.82	90.64	77.61	84.02
TrialCNN / Light	83.79	89.98	75.70	83.16
TrialCNN / Weather	82.62	86.70	77.98	82.44
ResNet-18 / Light	82.45	87.75	76.99	82.40
TrialCNN / Generalist	81.35	87.88	77.73	82.32
TrialCNN / Fog	83.28	87.82	75.33	82.14
ResNet-18 / Fog	82.31	89.19	74.08	81.86
TrialCNN / Noise	79.50	83.53	78.73	80.58
TrialCNN / Defocus	80.00	84.67	75.64	80.10

Full results: https://github.com/aswadekarb24-code/Machine_Shikavni

TABLE V: Ensemble-32 Mean Accuracy (%) and Weighted F1 per Corruption Type, averaged over the three datasets and all 5 severity levels.

Corruption	Accuracy (%)	Weighted F1
Zoom Blur	98.65	0.9862
Pixelate	98.42	0.9841
JPEG Compression	97.81	0.9781
Shot Noise	96.61	0.9658
Brightness	96.56	0.9658
Impulse Noise	93.89	0.9391
Gaussian Noise	93.72	0.9375
Elastic Transform	92.64	0.9268
Frost	87.82	0.8811
Snow	82.14	0.8216
Contrast	81.76	0.8399
Defocus Blur	78.19	0.7890
Motion Blur	75.80	0.7584
Fog	58.48	0.6114
Mean	**88.04**	**0.8846**

inputs. The blind setting (no corruption detector) is a deliberate design choice; a corruption-aware router could further improve results (see Section VII).

Fog as the persistent outlier. Fog at high severity fundamentally degrades spatial frequency content in ways that none of eight preprocessing transforms fully recovers. The local contrast normalization in `fog_tform` is effective at moderate severities but fails at the most extreme fog levels (severity 5). This is consistent with results in general image restoration literature where physical fog models require depth estimation beyond the scope of a classification pipeline.

VII. Limitations and Future Work

Several limitations bound the scope of our conclusions:

No corruption-aware training. All models are trained on clean data; evaluating corruption-augmented training (mixed or probabilistic corruption injection during training) is a natural next step and may close the residual robustness gap, particularly for fog and motion blur.

Blind ensemble. The current ensemble runs all 32 members on every input regardless of the actual corruption type. A lightweight corruption classifier could route inputs to the relevant specialist members and likely reduce the performance gap for fog and motion corruptions.

Dropped specialist variants. The `blur_tform` and `distort_tform` members were dropped after a sweep because they did not improve overall accuracy; their checkpoints are retained on disk. Revisiting them with architecture-specific hyperparameter tuning, or using them in a corruption-routed setting, remains an open direction.

Grayscale discards color cues. The robust preprocessing shield discards color information that could be informative for certain sign categories (e.g., red prohibition vs. blue mandatory signs). An ablation isolating grayscale vs. color preprocessing for individual classes is left for future work.

From-scratch training only. All experiments use models trained from scratch. Repeating the comparison with ImageNet-pretrained initialization would clarify whether the robustness advantage of the compact CNN is specific to the from-scratch, low-resolution regime or persists more generally.

VIII. Conclusion

We presented a three-phase progressive robustness study for traffic sign classification under 14 environmental and digital

corruption types. Starting from an RGB augmentation baseline (Phase 1), we introduced a Robust Preprocessing Shield – grayscale + median filter + CLAHE – that improved mean corrupted accuracy for TrialCNN by 5.3 percentage points at no architectural cost (Phase 2). We then developed a corruption-specialist ensemble framework of 8 preprocessing variants $\times$ 4 architectures $=$ 32 specialist models combined via confidence-weighted fusion (Phase 3), yielding a final **Ensemble-32 with 88.0% mean corrupted accuracy** and near-perfect clean accuracy ($>$97.8% across all datasets), outperforming the best single specialist model by 2.6 pp.

The central message of this work is that *targeted preprocessing diversity is a more effective and efficient robustness lever than raw architectural capacity for small-resolution traffic sign recognition*: matching each model's input transform to its expected corruption family, and combining the specialists via confidence weighting, extracts corruptionrobustness substantially beyond what any single model – compact or large – achieves alone.

All code, checkpoints, and full result tables are available at: https://github.com/aswadekarb24-code/Machine_Shikavni

ACKNOWLEDGMENT

We thank our project mentor for guidance and feedback throughout the evaluation process that shaped this study.

REFERENCES

[1] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "Man vs. computer: Benchmarking machine learning algorithms for traffic sign recognition," *Neural Networks*, vol. 32, pp. 323–332, 2012.

[2] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. ICLR*, 2019.

[3] J. Zhang, M. Huang, X. Jin, and X. Li, "A real-time Chinese traffic sign detection algorithm based on modified YOLOv2," *Algorithms*, vol. 10, no. 4, p. 127, 2017.

[4] D. S. Sarwatt, F. Kulwa, J. Ding, and H. Ning, "Adapting image classification adversarial detection methods for traffic sign classification in autonomous vehicles: A comparative study," *IEEE Trans. Intell. Transp. Syst.*, 2024.

[5] R. Timofte, K. Zimmermann, and L. Van Gool, "Multi-view traffic sign detection, recognition, and 3D localisation," in *Proc. IEEE WACV*, 2009.

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[7] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE CVPR*, 2018, pp. 4510–4520.

[8] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.

[9] T. G. Dietterich, "Ensemble methods in machine learning," in *Proc. MCS*, 2000, pp. 1–15.

[10] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991.